
UNIVERSITY OF CAPE TOWN
ECO4122Z
EXAMINATION QUESTION

INFORMATION TO CANDIDATES

- i. Remember this section contributes 10% of the final examination mark.
- ii. Marks indicated below are for guidance purposes; in terms of relative weight of each section
- ii. Your submission should be within 45 minutes after completing the section under the Test/Quizzes Tab.

Question

Martha Vilakazi is the Managing Director of Stationery Mint Company, a South African-based stationery manufacturing and distribution company whose sales are entirely domestic and whose shares are listed on the Johannesburg Stock Exchange. The following are additional facts concerning her current situation:

- Vilakazi is single, aged 58. She has no immediate family, no debts, and does not own a residence. She is in excellent health and covered by Stationery Mint-paid health insurance that continues after her expected retirement at age 65.
- Her base salary of R900,000/year, inflation-protected, is sufficient to support her present lifestyle but can no longer generate any excess for savings.
- She has R3,000,000 of savings from prior years held in the form of short-term instruments.
- Stationery Mint rewards key employees through a generous stock-bonus incentive plan but provides no pension plan and pays no dividend.
- Vilakazi's incentive plan participation has resulted in her ownership of Stationery Mint stock worth R8 million (current market value). The stock, received tax-free but subject to tax at a 40% rate (on entire proceeds) if sold, is expected to be held at least until her retirement.
- Her present level of spending and the current annual inflation rate of 4% are expected to continue after her retirement.
- Vilakazi is taxed at 41% on all salary, investment income, and realized capital gains. Assume her composite tax rate will continue at this level indefinitely. Vilakazi's orientation is patient, careful, and conservative in all things. She has stated that an

annual after-tax real total return of 3% would be completely acceptable to her if it was achieved in a context where an investment portfolio created from her accumulated savings was not subject to a decline of more than 10% in nominal terms in any given 12-month period. To obtain the benefits of professional assistance, she has approached your company for investment advisory.

1. Formulate the risk-return objectives and constraints portion of her investment policy statement addressing each of the following:
 - i. Return Requirement.
 - ii. Risk Tolerance.
 - iii. Time horizon
 - iv. Liquidity requirements
 - v. Tax concerns
 - vi. Unique circumstances

[15 marks]

2. You have tasked five(5) of your newly recruited interns, Jamal, Yongxin, Nigel, Dylan and Yanga to present their proposed asset allocations for Vilakazi's portfolio, under the assumption that only the current yield portion of projected total return (comprised of both investment income and realized capital gains) is taxable to Vilakazi and that the municipal bond income is entirely tax-exempt (see the Table 1; below).

Table 1: **Proposed asset allocation from the Interns**

Asset Class	Projected Return	Expected Std Dev.	Jamal	Yongxin	Nigel	Dylan	Yanga
Cash equivalents	4,50%	2,5%	10%	20%	25%	5%	10%
Corporate bonds	6	11	0	25	0	0	0
Municipal bonds	7.2	10.8	40	0	30	0	30
Large-cap stocks	13	17	20	15	35	25	5
Small-Medium cap stocks	15	21	10	10	0	15	5
International stocks	15	21	10	10	0	15	10
Real estate investment trusts (REITs)	10	15	10	10	10	25	35
Venture capital	26	64	0	10	0	15	5
Total			100	100	100	100	100
Summary Projections							
Total return			9.90%	11.00%	8.80%	14.40%	10.30%
After-tax total return			7.40%	7.20%	6.50%	9.40%	7.40%
Standard deviation			9.40%	12.40%	8.50%	18.10%	10.10%
Sharpe ratio			0.4681	0.4435	–	0.4917	0.4752

- (a) Determine which of the asset allocations meets or exceeds Vilakazi's stated return objective. [2 marks]
 - (b) Determine any two asset allocations that meet Vilakazi's risk tolerance criterion. [2 marks]
 - (c) Assuming a risk-free rate of 5.5%, calculate the Sharpe ratio for Nigel's Asset Allocation. [1 mark]
 - (d) Recommend and justify the one asset allocation that would be most suitable for Vilakazi's savings portfolio. [5 marks]
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